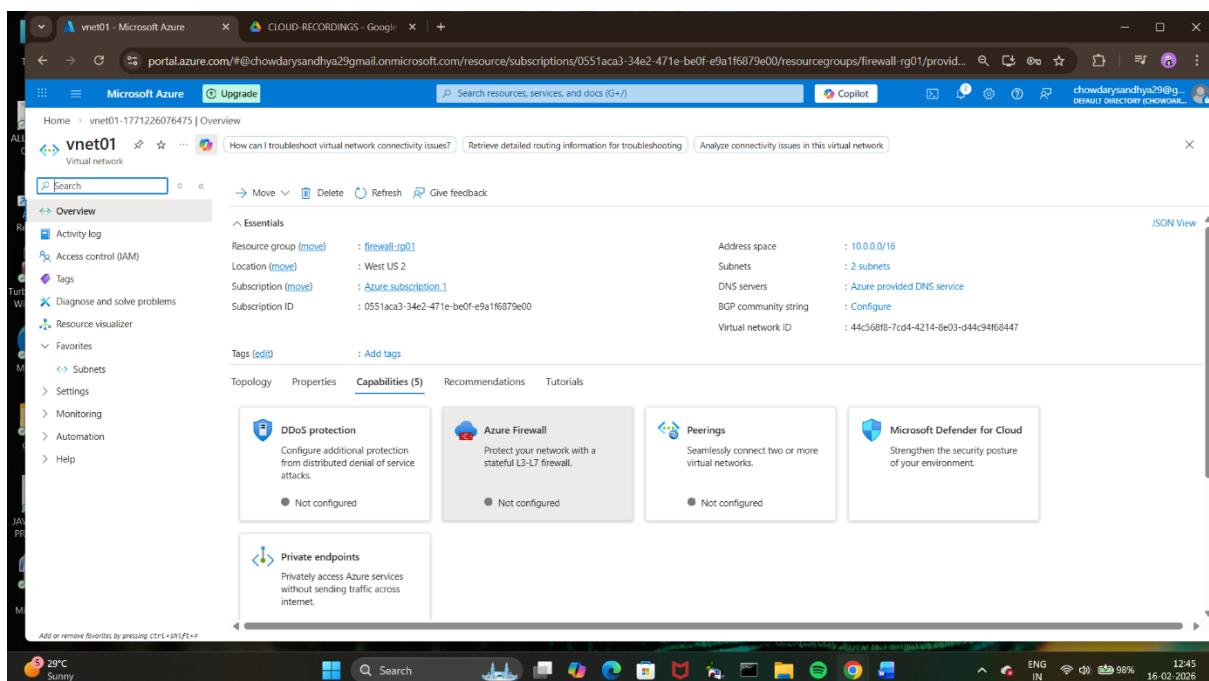
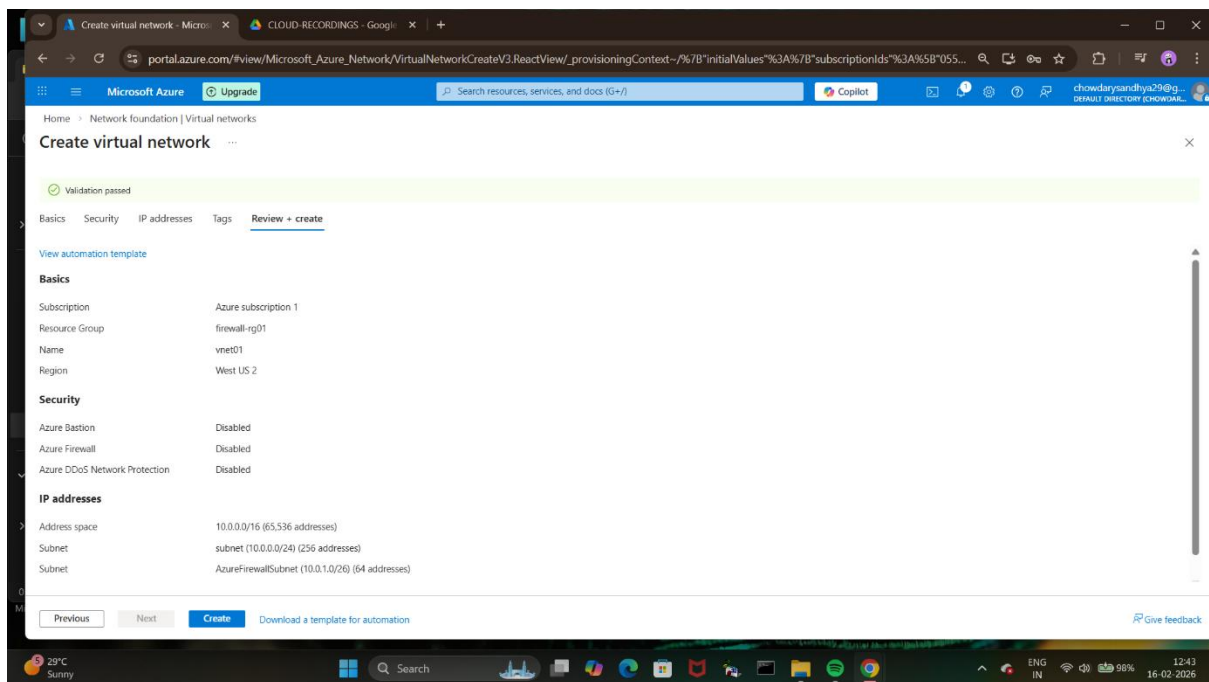
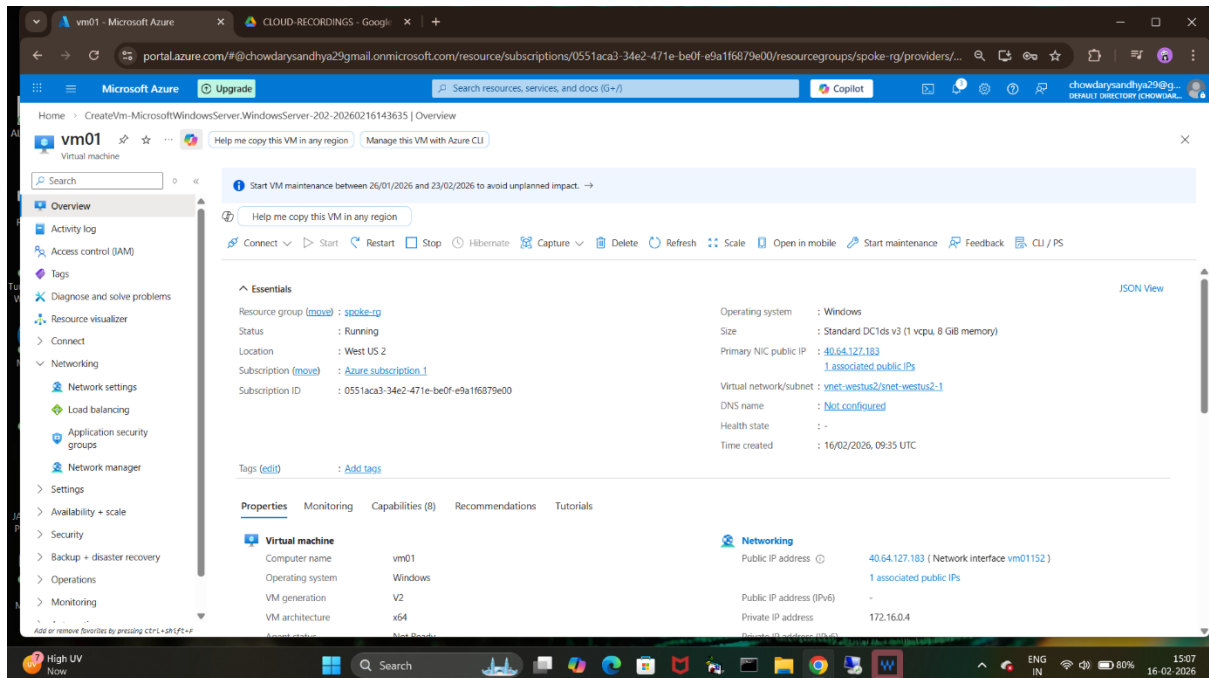


Azure firewall

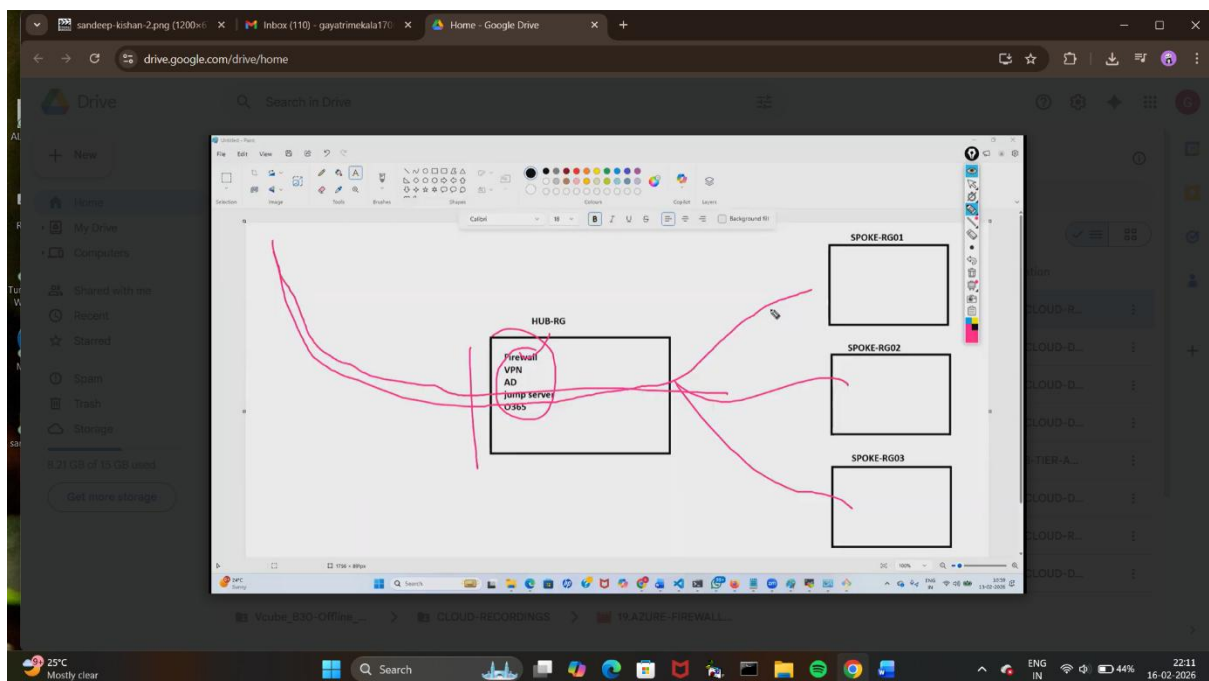
- Create Rg (2)
- Create firewall with 2 subnets 1.default (subnet)
2.firewall (firewall subnet) dedicated subnet
- Next create spokenVM with spokeVnet and its subnet.
- Create virtual network



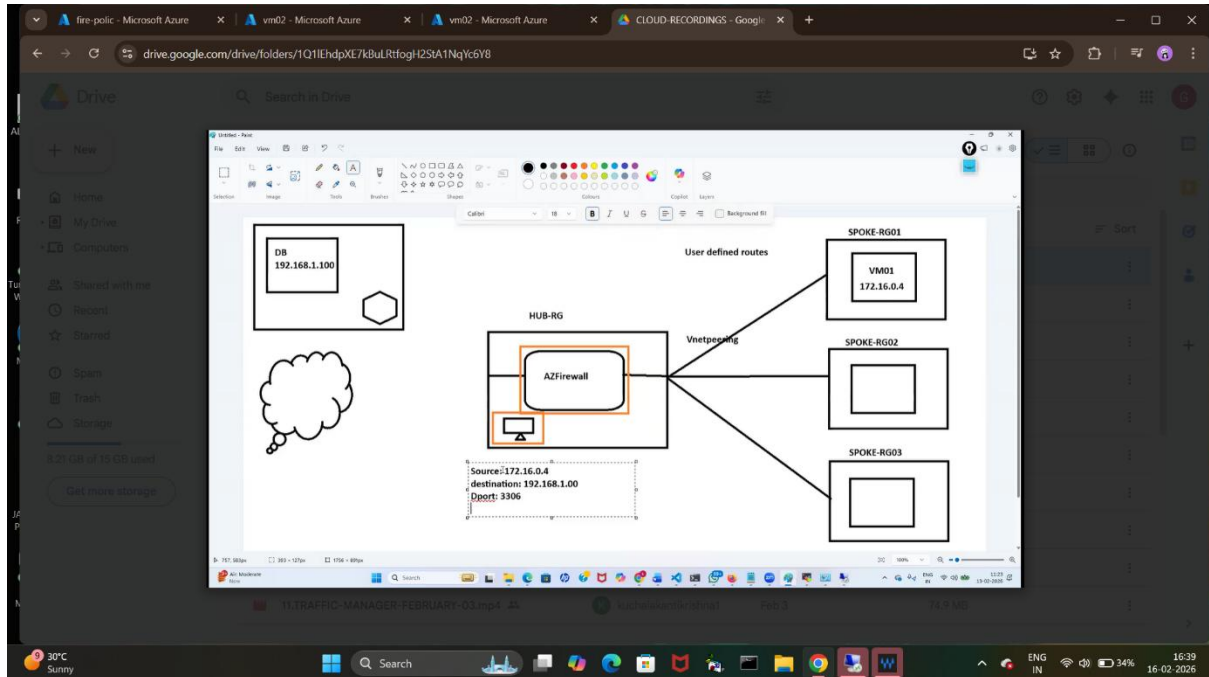
- Now, create a firewall
- Next create spoke Vm in same region and give its subnet.
- NOW CREATE VIRTUAL MACHINE



HUB AND SPOKE ARCHITETURE

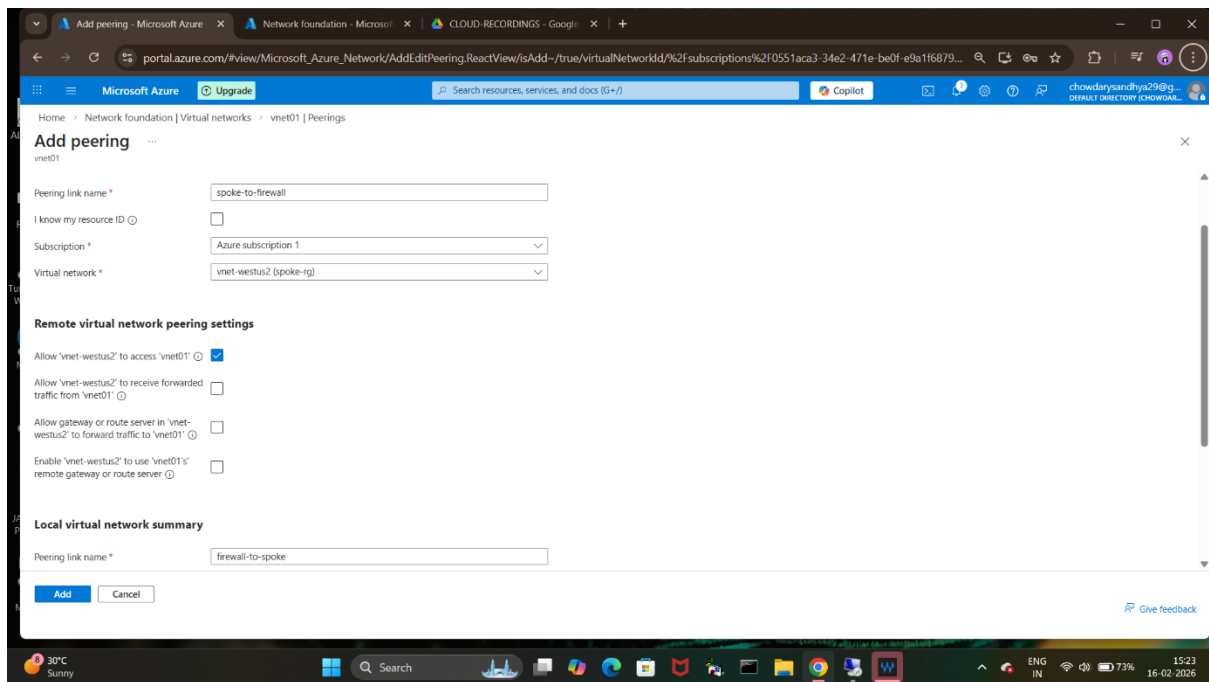


g

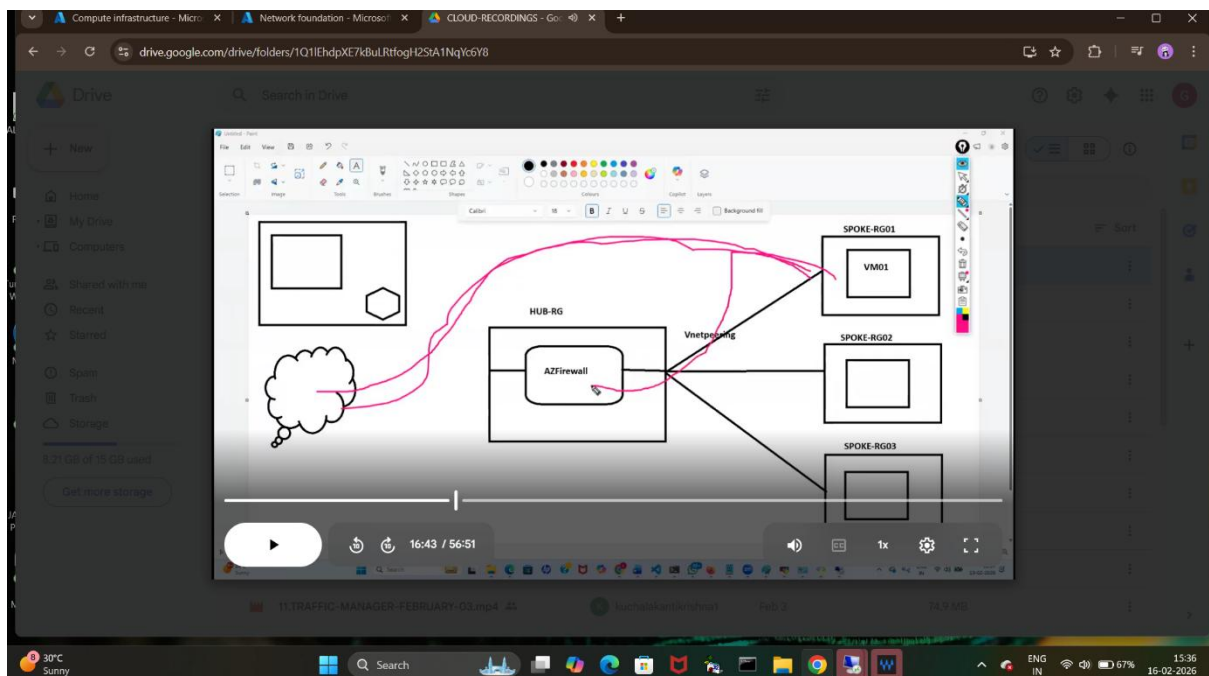


- ✓ Machine behind the firewall, by default there will be denied rule we can't access the machines
- ✓ From public even though machines have public Ip once the machine is behind the firewall by default traffic will be blocked
- ✓ For now, we created firewall and spokevm (windows)
- ✓ Let's log into spoke vm (pip)
- ✓ Next lets configure VNET PEERING from firewall vnet & spoke vnet
- ✓ We see two diff Vnet for firewall and spoke below

✓ Choose any vnet for peering

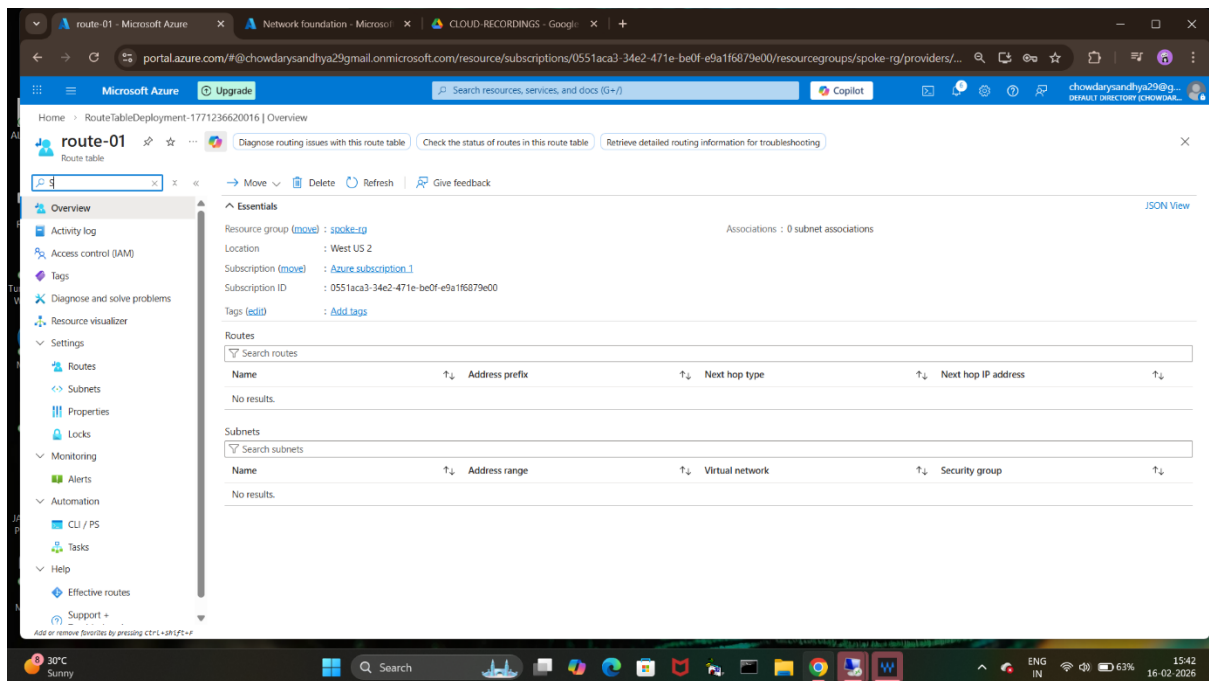


- ✓ After vnet peering we can still connect to spoke vm because the traffic is going through outside and connecting to spokeVM .



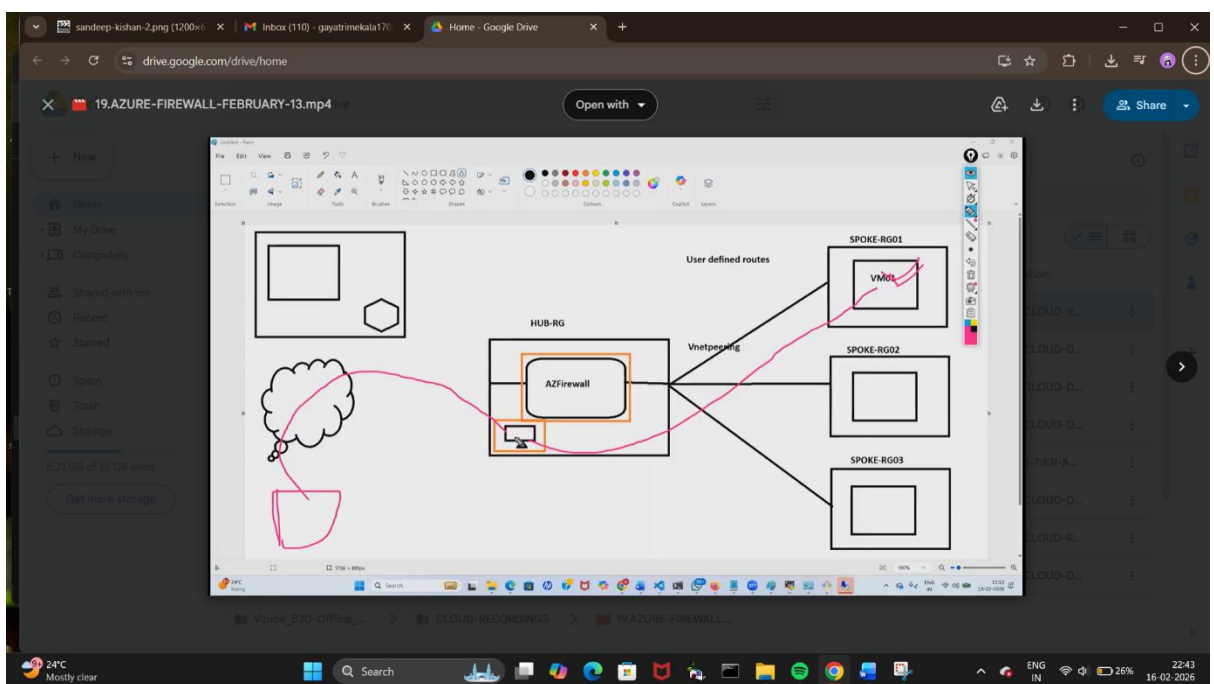
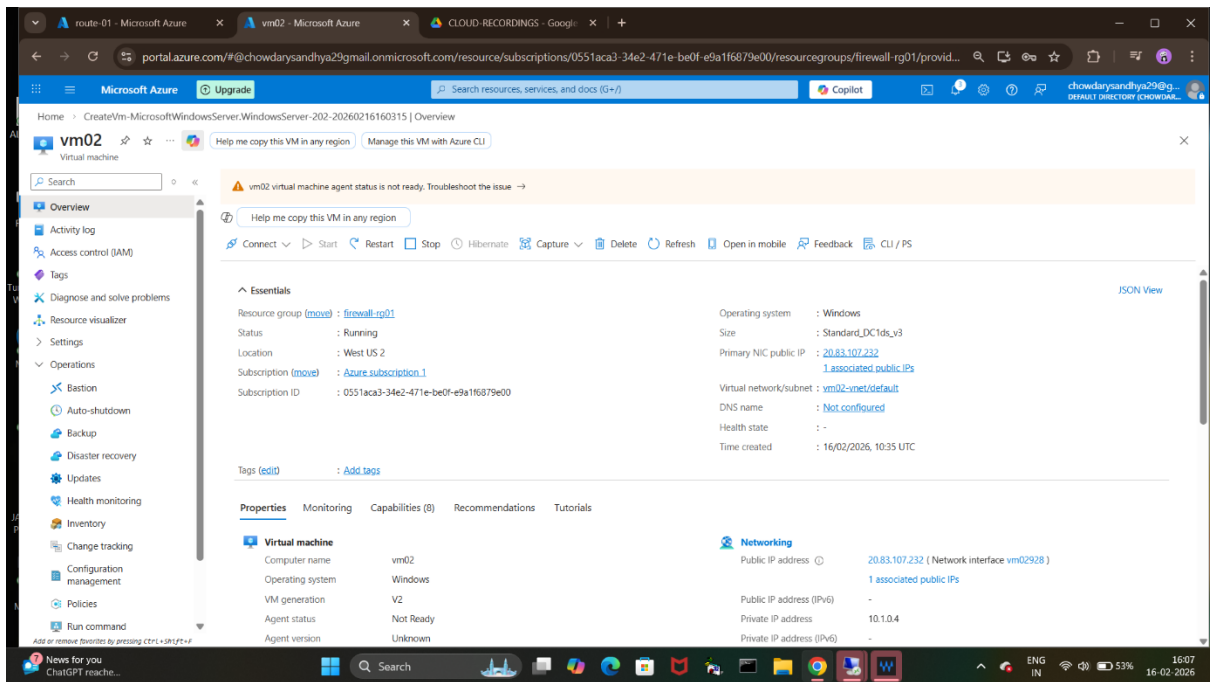
- ✓ Lets Redirect traffic to firewall
- ✓ For that use should right route table (user defined routes)
- ✓ Search for route tables and click on create

CREATE ROUTE TABLE



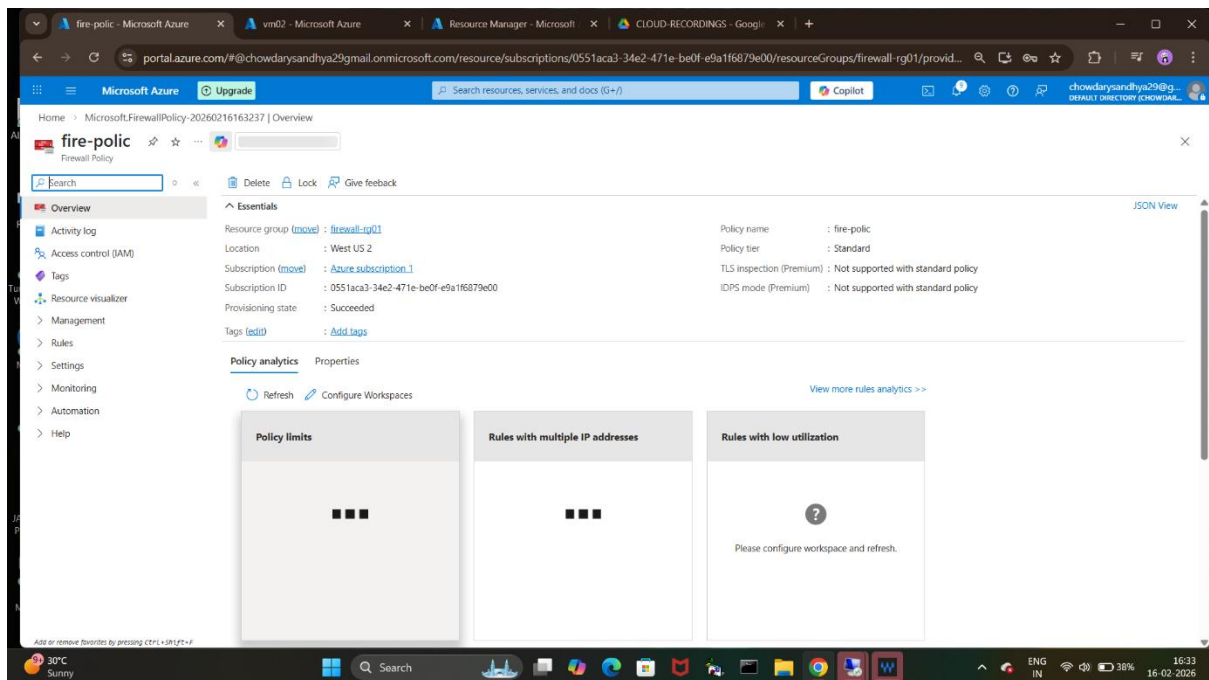
Here, above creation of route table we have to add this –

- 0.0.0.0/0 is any above. Copy below firewall private Ip below
- Now from spoke VM whatever the traffic it should hit firewall first
- Now next we write the route rule it should be attached to subnet
- We have defined routing, so the traffic coming from inside it should use azure firewall only, the traffic will flow from firewall only.
- After adding the rules to subnet automatically spoke VM which is logged in earlier it will disconnect, because traffic diverted from firewall.
- Next, we have to connect RDP with the Ip (VM)
- Next create another VM using subnet in firewall Vnet
- This VM just to know what's happening in spoke VM .
- To access this from my LAPTOP---VM---SPOKEVM



- Let's create VM (windows) on HUB-RG (firewall Vnet)
- Now connect to this VM using public Ip(72.146.0.169) on RDC—the Ip address is what we are had that only not a particular
- After connecting to VM, open RDC in it and connect to spoke VM internally by using Spoke VM private IP (172.18.0.4)
- We can't access anything because internet is not allowed and rules not match on firewall and this is behind the firewall.
 - ✓ Let's talk about rules (follow below steps)

Create an azure firewall policy

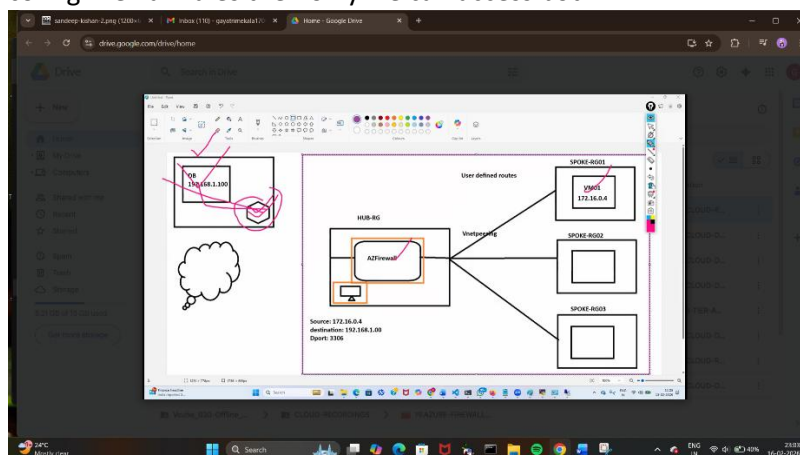


Default policy

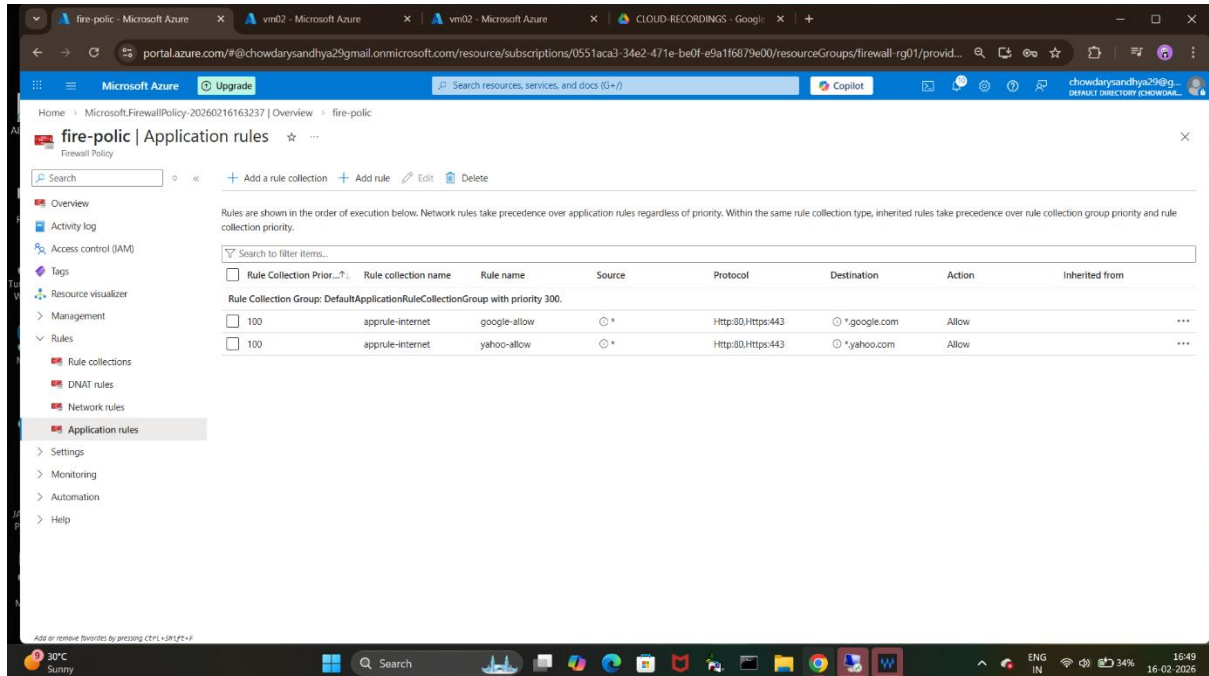
- ✓ APPLICATION RULES
- ✓ DNAT RULES
- ✓ NETWORK RULES
- ✓ **Network rule if want to allow Ip to Ip then we use it.**
- ✓ We don't use domain names here

• Below you can see example which we used in lab.

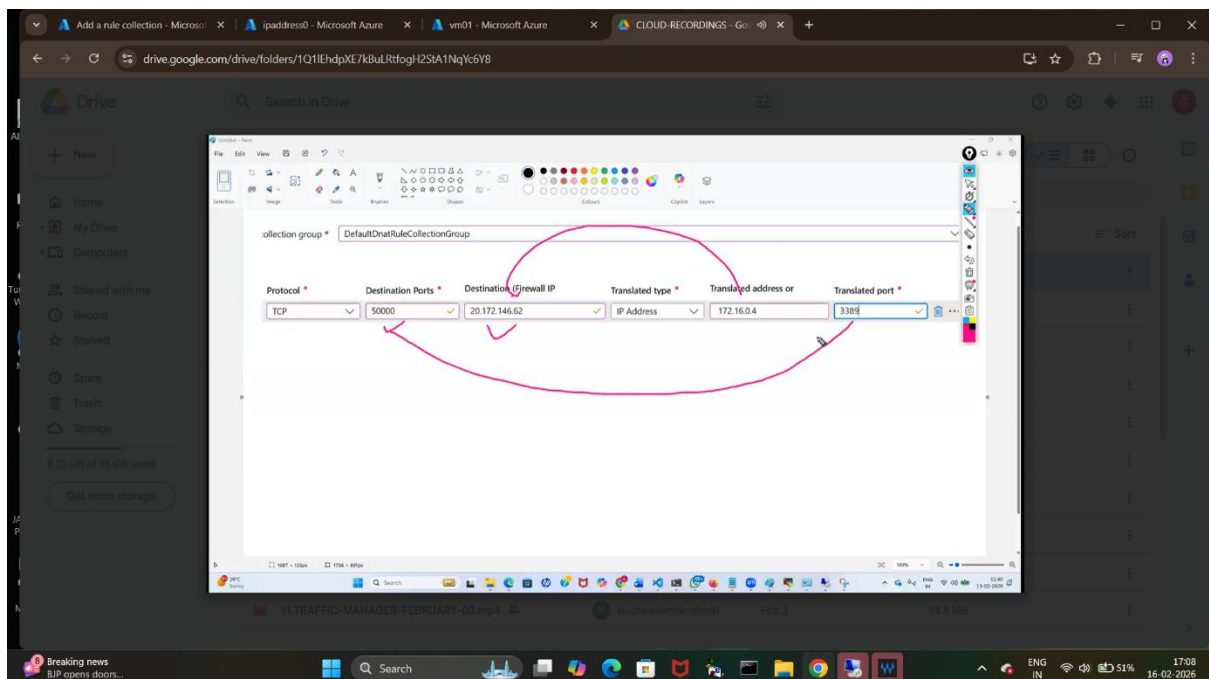
- ✓ App and DB in different datacentres and they both have firewalls and both should config firewall rules then only we can access both.



- ❖ Application rules
- ❖ Associate this default policy to your firewall
- ❖ Go to firewall –Migrate to firewall policy –attach an existing firewall policy below

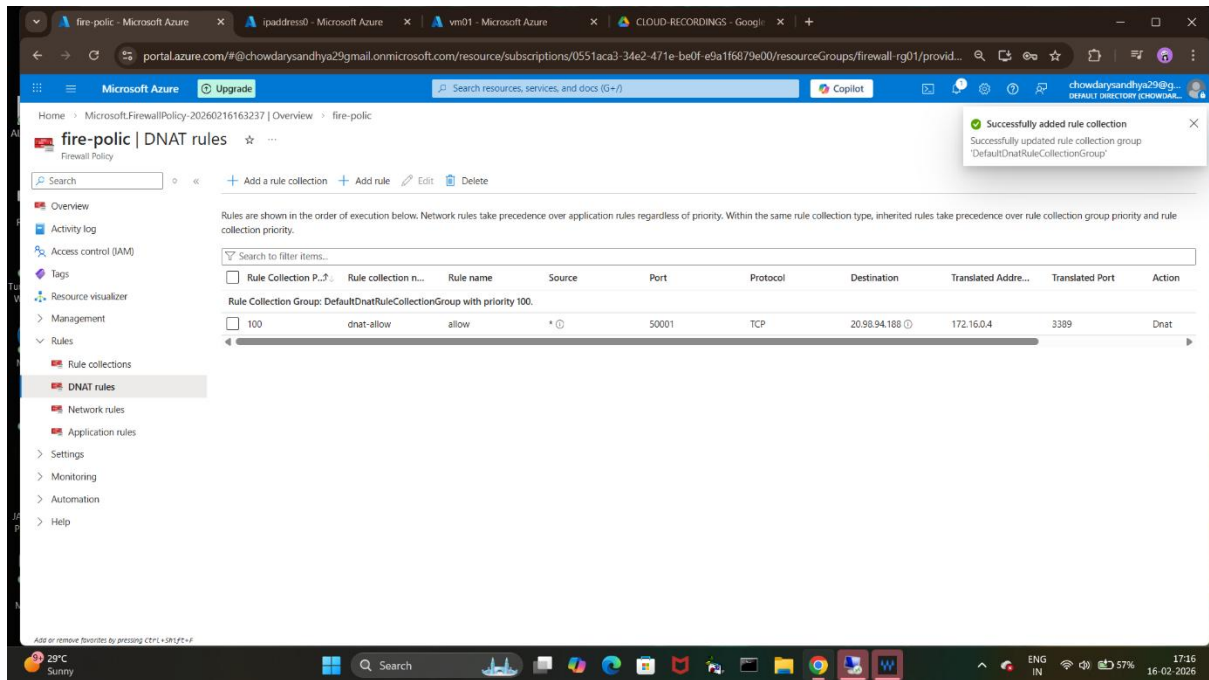


- ❖ Select above policy and save

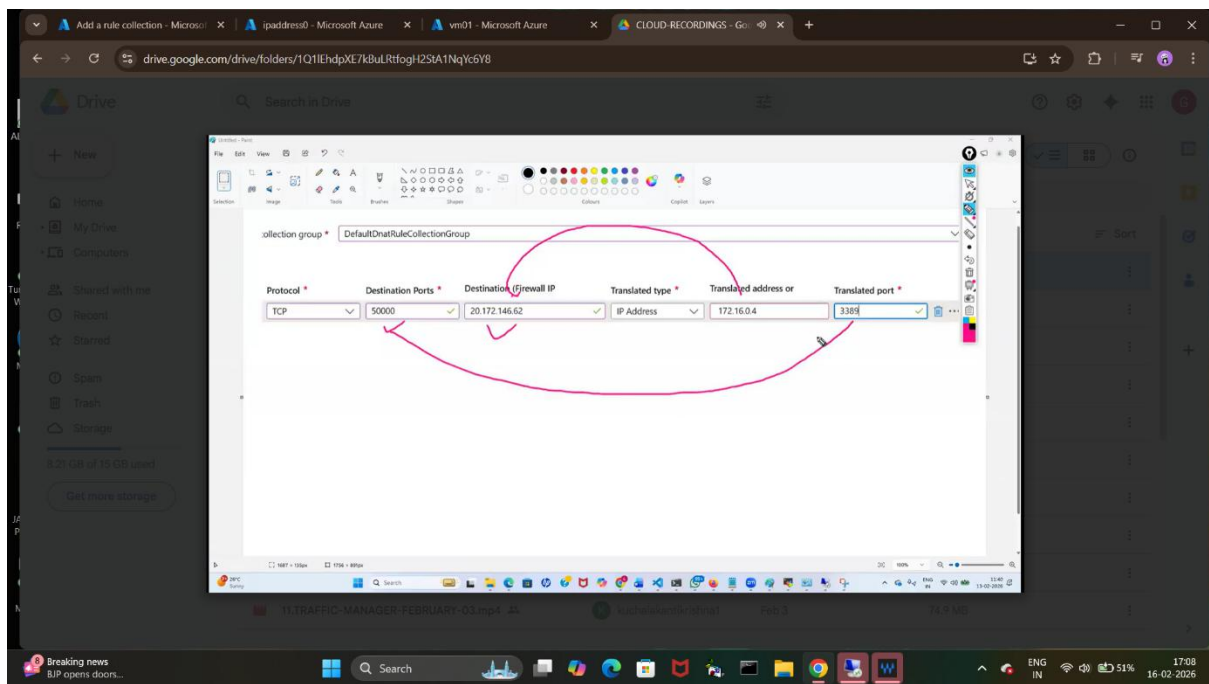


- DNA rules
- ✓ Below I want to connect from internet to backend machine
- ✓ efeeconnect because the machine is behind the firewall .
- ✓ But in this case, we can cannot by use firewall Public Ip

- ✓ Outside traffic hits to firewall public Ip from that traffic will go to backend machine, here backend private Ip masked by firewall.
- ✓ Firewall Public Ip (172.213.217.109)—Backend spoke VM private Ip (172.18.0.4)
- ✓ Same with ports 5000—3389
- ✓ Translating with public—private and from port—port security to machine , we are just mapping with one to one
- ✓ We can choose servers; on those we can add this mapping at in emergency situations then access backend machines.



- ❖ There is another option DNATting—with private Ips only by using VPS , we can login into the azure infrastructure to one of the gem servers from there we are going to connecting .



Copy firewall public Ip :50000

- ✓ By this we can connect to backend SpokeVM
- Then connect to the RDP

